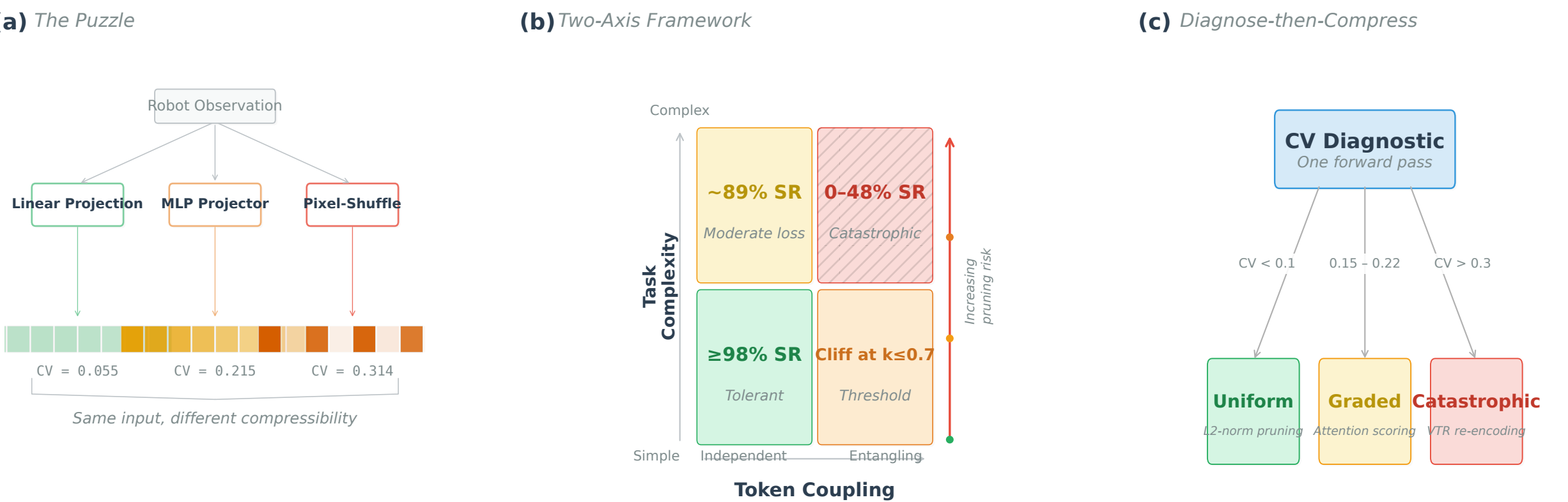
(a) The Puzzle
Robot Observation
Linear Projection
MLP Projector
Pixel-Shuffle
CV = 0.055
CV = 0.215
CV = 0.314
Same input, different compressibility
(b) Two-Axis Framework
Complex
Task Complexity
Simple
~89% SR
Moderate loss
0–48% SR
Catastrophic
≥98% SR
Tolerant
Cliff at k≤0.7
Threshold
Independent
Entangling
Token Coupling
Increasing pruning risk
(c) Diagnose-then-Compress
CV Diagnostic
One forward pass
CV < 0.1
0.15 – 0.22
CV > 0.3
Uniform
L2-norm pruning
Graded
Attention scoring
Catastrophic
VTR re-encoding